

Performance Testing

Date	28 Jun 2026
Team ID	XXXXXX
Project Name	Smart Crop
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust & Postman API Automation Suite
Type of Testing	Load Testing and Spike Testing
Target Module / API	`/api/v1/recommend` (Scenario 1 Core Optimization Endpoint)
Test Environment	Cloud Hosted AWS Sandbox Server Environment
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Scenario 1 Load Baseline: Simulate steady multi-farmer soil profile data inputs (\$N, P, K, pH\$, temperature, humidity, rainfall).	1,000	300	Smooth execution, zero error flags.

2	Scenario 2 Peak Window Spike: Simulate heavy concurrent crop suitability evaluation and climate compliance lookups.	5,000	1201	Average response latency stays under 2 sec.
3	Scenario 3 Researcher Batch Load: Aggregated macro query analysis requests run over regional spatial data arrays.	200	180	System returns CSV/ PDF data logs accurately.

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	0.45 seconds	PASS	
2	Response Time (Max)	< 5 seconds	1.85 seconds	PASS	
3	Throughput (Req/sec)	> 500 req/sec	850 req/sec	PASS	
4	Error Rate	< 1%	0.02%	PASS	
5	CPU Utilization	< 80%	42%	PASS	
6	Memory Utilization	< 80%	56%	PASS	

Step 4: Observations & Analysis

KEY FINDINGS:

The predictive engine performed exceptionally under simulated steady-state conditions, processing the 7 target metrics (N, P, K,pH, temperature, humidity, rainfall) with an average latency well below the target thresholds.

BOTTLENECKS IDENTIFIED:

A minor latency hike was discovered during spike tests on multi-variable scenario lookups, which was traced to inefficient relational database query joins when pulling extensive climate baseline profiles.

OPTIMIZATION STEPS TAKEN:

We implemented database indexing for primary crop variables and set up an in-memory caching mechanism to store unchanging weather historical models, lowering peak latency.